

Milan Barot

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SUMMARY

Data Engineer with 6 years of experience specializing in healthcare data engineering, payer/TPA systems, ETL, and legacy-to-modern platform migration. Experienced with HealthRules Payer (HRP), claims, eligibility, accumulator, provider, and financial data, building and optimizing SQL-based data pipelines and reporting solutions across on-premises and cloud environments. Proven track record of translating complex healthcare business requirements into scalable data solutions while ensuring data accuracy, reliability, and production readiness. Currently expanding into modern data engineering with Apache Spark, Databricks, and lakehouse architectures, with hands-on experience building end-to-end data platforms using Medallion architecture, Delta Lake, and AI-powered analytics.

TECHNICAL SKILLS

Data Engineering: SQL, Python, PySpark, Apache Spark, SSIS, Azure Data Factory, ETL

Cloud & Lakehouse: Azure, Databricks, Delta Lake, Medallion Architecture

Databases/Data Warehouse: SQL Server, Azure SQL, PostgreSQL, DuckDB

BI & Reporting: Power BI, SSRS

Data Integration: EDI 834/837, REST APIs, FTP, ActiveBatch, SQL Server Agent

Workflow & DevOps: Airflow, Dagster, Docker, Git, Jenkins, Jira

Application & AI: FastAPI, React, Electron, ChromaDB, OpenAI API, Groq API

WORK EXPERIENCE

SQL Server Developer / Data Engineer

Jan 2026 – Present

AmeriBen (UST Global — Implementation Partner)

Remote

- Contributed to the migration of healthcare reporting from a legacy **GBAS** platform to an **HRP-based ODS** architecture, supporting the onboarding of self-funded employer groups and transition of financial, utilization, claims, census, and clinical reporting to the new platform.
- Developed and maintained complex **SQL Server** reporting logic, stored procedures, merge scripts, and **ETL** workflows supporting reports for large claims, financial summaries, contract savings, utilization, lag analysis, and level-of-coverage census across legacy and modernized data sources.
- Built consolidated reporting datasets spanning both GBAS and ODS during the platform transition, reconciling fundamentally different schemas and business structures to ensure consistent reporting for clients operating across legacy and HRP-derived data.
- Investigated data discrepancies through the HPS → ODS transformation pipeline, analyzing source HRP data, mappings, ETL logic, and downstream reporting objects; implemented corrective changes to ensure accurate representation of migrated healthcare data.
- Translated complex payer/TPA business requirements into production-grade SQL and reporting solutions, resolving ETL issues, SQL performance bottlenecks, cross-database dependencies, and data-quality issues across development, testing, and production environments.
- Collaborated with business stakeholders and HRP teams to understand healthcare reporting requirements, validate data mappings and business rules, and deliver technical solutions aligned with project requirements and migration milestones.
- **Tech Stack:** SQL Server 2022, Azure SQL, SSRS, SSIS, ETL, HealthRules Payer (HRP).

Data Engineer

Sep 2022 – Oct 2025

Blue Cross Blue Shield Arizona (Mastek — Implementation Partner)

Phoenix, AZ

- Designed and implemented end-to-end **ETL** pipelines using **SSIS** and **SQL Server** to process eligibility, claims, accumulator, and value-based care (VBC) files, ensuring compliance with business requirements and vendor-specific standards. Collaborated with vendors such as OptumRx, HealthEquity, Equality Health, and others to deliver high-quality data solutions.
- Led offshore engineering resources across multiple ETL initiatives, managing task assignments, technical dependencies, and delivery timelines.
- Utilized **Azure Data Factory (ADF)** to design and orchestrate scalable data integration pipelines for migrating legacy member and claims data to **HealthRules Payer (HRP)** on Azure Cloud.
- Leveraged **Azure SQL Database** for staging, transformation, and persistent storage of high-volume healthcare data. Configured ADF linked services, pipelines, and triggers for automated, scheduled data ingestion from on-premises SQL Server to Azure.

- Rebuilt critical ETL pipelines using SSIS and SQL Server after the **CHC system hack (2024)**, restoring high-priority data workflows under tight deadlines with minimal downtime.
- Developed complex data extracts that combined on-prem warehouse data with select cloud-based sources, ensuring consistency, accuracy, and seamless integration across hybrid environments, reducing execution time by **25%** on average across the board.
- Collaborated closely with **Deloitte** on the HealthEquity project to develop eligibility and claims files and automated it end to end. Responsibilities included gathering business requirements, creating technical documentation, optimizing ETL pipelines, conducting performance testing, and ensuring smooth deployment into production environments.
- Automated data workflows by integrating SQL Server Agent jobs with **ActiveBatch**, setting up FTP connectivity with strict naming conventions and cadence-driven schedules for file delivery to vendors.
- Developed a consolidated **EDI framework for 834 and 837** enrollments, claims and dental, using three core SSIS packages to automate the generation and delivery of over **35 distinct file types** to CMS. This architecture ensured secure data exchange and audit readiness while significantly reducing the manual maintenance and operational overhead previously required.
- Provided active production support for mission-critical CMS data exchanges, conducting over **25 successful production deployments** and managing emergency releases while maintaining compliance with federal reporting deadlines.
- Led the rapid resolution of critical production failures, ensuring minimal data loss during high-volume enrollment periods where direct connectivity with CMS was required.
- Partnered with business stakeholders to gather requirements for new projects, conducting detailed assessments of complexity, technical effort, development timelines, and warranty periods. Worked with Mastek to draft SOWs for approval by Blue Cross Blue Shield, ensuring alignment with business objectives.
- Participated in Agile sprint meetings to track progress on deliverables, resolve blockers, and ensure timely completion of project milestones across multiple initiatives.
- **Tech Stack:** Azure Data Factory (ADF), Azure SQL Database, C#, SQL Server, SSIS, SSRS, ActiveBatch, SQL Server Agent, FTP, EDI (834, 837), HealthRules Payer (HRP), Agile +4.

Data Engineer

Feb 2021 – Sep 2022

CBRE

Dallas, TX

- Played a key role on the Data Engineering & Reporting team during the COVID-19 pandemic, delivering real-time data pipelines and analytics that supported CBRE's Advisory business unit across global commercial real estate portfolios.
- Designed, developed, and maintained robust ETL pipelines using **Python, SQL, and PostgreSQL** via CBRE's internal ETL framework (Lymbic). Translated complex business logic from real estate analysts into SQL/Python transformations for property usage, tenant risk, and leasing performance.
- Ingested diverse external data sources such as Google Mobility Reports and BREAM energy efficiency assessments, transforming them into structured datasets in Parquet and CSV formats. These outputs fed into PowerBI dashboards used by leadership to assess region-wise COVID impacts on commercial spaces.
- Applied data profiling and transformation using **Pandas** and **PySpark** for efficient in-memory processing of large CSV/Parquet files, enabling faster QA cycles and stakeholder feedback loops.
- Supported **Databricks** orchestration workflows — monitoring job execution, rerunning failed tasks, and assisting engineers with job recovery — ensuring daily and weekly batch pipelines remained consistent and reliable during volatile COVID data conditions.
- Worked in an Agile delivery environment, actively collaborating with analysts, product managers, and backend engineers in weekly sprint meetings to prioritize features, track bugs, and refine pipeline outputs based on stakeholder needs.
- Enabled high-impact real estate decisions by producing clean, trusted data for dashboards that revealed critical trends such as building underutilization, shifting tenant behaviors, and geographic leasing risk. These insights directly influenced CBRE's portfolio restructuring strategies and client advisory recommendations during the pandemic.
- **Tech Stack:** Python, SQL, PostgreSQL, Power BI, Airflow, Pandas, Parquet, Agile.

ETL / SQL Server Developer

Oct 2019 – Aug 2020

AmNet Services Inc.

Fairfield, NJ

- Developed **SSIS** packages to automate extraction and ingestion of network test data from Excel, flat files, and MS Access sources into a SQL Server staging environment.
- Designed and implemented a dimensional data warehouse model using **ERwin Data Modeler** to support analytical queries on network KPIs.

- Wrote complex **T-SQL** stored procedures for data cleansing, aggregation, and validation to ensure accuracy and consistency of telecom metrics.
- Created and maintained **SSRS** reports and PowerBI dashboards to visualize signal strength, call drop rates, and E911 test call success metrics.
- Automated SQL Server Agent jobs to schedule SSIS package executions and report generation, ensuring daily data processing without manual intervention.
- Collaborated with radio engineers, business analysts, and compliance teams in Agile sprints to gather requirements and iteratively improve data solutions.
- Conducted rigorous testing of ETL workflows and reports to guarantee data quality and system performance at scale.
- **Tech Stack:** SSIS/SSRS, T-SQL, Power BI, Agile.

SELECTED PROJECTS

Kalshi-Prediction-Engine — AI-Powered Intelligence Lake-House | *PySpark, Delta Lake, ChromaDB, Groq, React*

- Architected a full-stack prediction market intelligence platform using a **Medallion (Bronze/Silver/Gold) data lakehouse**, processing 6 global news feeds, GDELT event streams, and Kalshi prediction market data through **PySpark ETL** with **Delta Lake ACID** guarantees.
- Engineered a **RAG-powered inference engine** that embeds Silver-layer data into ChromaDB (MiniLM-L6-v2), performs cascading time-windowed semantic search, and synthesizes bull/bear/verdict intelligence briefs via Groq (Llama 3.3 70B) — including automated “Ghost Pump” detection for unsupported price movements.
- Built a **React Human-in-the-Loop trade terminal** backed by a FastAPI bridge server, surfacing AI-generated intelligence briefs with Kelly Criterion bet-sizing and a Safe Mode guard for risk-controlled live trade execution on Kalshi via RSA-signed API calls.
- Deployed a **10-container Docker Compose** production environment with a custom ETL orchestrator (watermark-based change detection, lock-file API coordination, once-daily settlement sweeps) running autonomously at \$0/month on local Windows infrastructure.
- Engineered a resilient, cost-efficient news ingestion pipeline across diverse political sources, leveraging RSS discovery, Trafilatura full-text extraction, and VADER sentiment analysis; optimized Kalshi series-first filtering to reduce API calls by approximately **98%**.

RealTime_Context_Engine — AI-Powered Windows Runtime Host | *Electron, FastAPI, OpenAI, RSA, SSE*

- Developed a dual-process desktop application utilizing an **Electron** frontend and a **Python/FastAPI** backend to provide real-time interview guidance.
- Integrated OpenAI GPT models via low-latency streaming (SSE) to deliver generative feedback with a <200ms TTFT (Time To First Token).
- Built a custom hardware-locked licensing system using **RSA digital signatures** and machine-specific HWIDs to secure distribution.
- Engineered a high-performance screen capture and audio processing pipeline for real-time context analysis during live sessions.
- Automated a full distribution pipeline using **PyInstaller** and **Electron Builder**, incorporating code signing and an automated update system.

EDUCATION

Washington University of Science & Technology
B.S. in Information Technology — Software Engineering

Vienna, VA
 Mar 2022

The City University of New York
B.A.S. in Computer Information Systems

New York, NY
 Dec 2019

CERTIFICATIONS

Databricks Certified Data Engineer Associate

Oct 2026